

Parameter estimation

$f(x; \theta)$ max. likelihood

$$f(\underline{\theta} | \underline{x}) = \frac{f(\underline{x} | \underline{\theta}) f(\underline{\theta})}{\int f(\underline{x}, \theta) d\theta} \quad \text{Bayesian view}$$

Maximum
a posteriori

① Bernoulli distribution

$X_1, X_2, \dots, X_n$ independent draws
from Bernoulli(p)

$$\forall i \quad X_i = \begin{cases} 1 & \text{w.p. } p \\ 0 & \text{w.p. } 1-p \end{cases}$$

$$P(X_i = 1) = p$$

$$P(X_i = 0) = 1-p$$

$$P(X_i = x_i) = p^{x_i} (1-p)^{1-x_i} \quad x_i \in \{0, 1\}$$

$$f(x_1, x_2, \dots, x_n; p) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i}$$

$$\therefore \log f = \sum_{i=1}^n x_i \log p + (1-x_i) \log(1-p)$$

$$\frac{\partial \log f}{\partial p} = 0 = \sum_{i=1}^n \frac{x_i}{p} + \sum_{i=1}^n \frac{(1-x_i)}{1-p} (-1) = 0$$

$$\frac{\sum_{i=1}^n x_i}{p} = \frac{n - \sum_{i=1}^n x_i}{1-p} \rightarrow \hat{p} = \frac{1}{n} \sum_{i=1}^n x_i$$

(2)

Poisson

$$X_i \sim \text{Poisson}(\lambda) \quad f(x_i; \lambda) = \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$$

$$f(\{x_i\}_{i=1}^n; \lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$$

$$\log f = \sum_{i=1}^n -\lambda + x_i \log \lambda - \log(x_i!)$$

$$\frac{\partial \log f}{\partial \lambda} = -n + \sum_{i=1}^n x_i / \lambda = 0$$

$$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n x_i$$

(3)

Gaussian $N(\mu, \sigma^2)$

$$f(x_i; \mu, \sigma^2) = \frac{e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}}{\sigma \sqrt{2\pi}}$$

$$f(\{x_i\}_{i=1}^n; \mu, \sigma^2) = \prod_{i=1}^n \frac{e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}}{\sigma \sqrt{2\pi}}$$

$$\log f = \sum_{i=1}^n \left[-\frac{(x_i - \mu)^2}{2\sigma^2} - \log \sigma - \log \sqrt{2\pi} \right]$$

$$\frac{\partial \log f}{\partial \mu} = \sum_{i=1}^n \frac{-1}{\cancel{2\sigma^2}} \cancel{2} (x_i - \mu) \cancel{(-1)} = 0$$

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\frac{\partial \log f}{\partial \sigma} = \sum_{i=1}^n \frac{-(x_i - \mu)^2}{2} \frac{(-2)}{\sigma^3} - \frac{n}{\sigma} = 0$$

$$\hat{\sigma} = \sqrt{\sum_{i=1}^n \frac{(x_i - \hat{\mu})^2}{n}}$$